Market Pulse — Development Session Changes

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1. Bug Fixes

Anomaly Detection — 0 Results (Root Cause: Insufficient Data)

The Risk/Anomaly tab showed 0 anomalies with "No rows available." because all detectors require a minimum of **5 days** of historical data to establish a z-score baseline. With only 4 scraped days, every detector was correctly returning empty — not a code bug, but a data constraint.

DataFrame `or` Ambiguity Errors (3 locations in `main.py`)

Multiple pipeline stages used Python's `or` operator as a None-fallback for DataFrames (e.g., `_run_ctx.get("df_full") or load_price_data()`). When a DataFrame was already stored in `_run_ctx`, Python evaluated `bool(DataFrame)` which raises `ValueError`. Fixed all 3 occurrences to use explicit `if key in _run_ctx` checks.

Affected stages: AI Narratives, Anomaly Detection, Historical Fetch

Forecast Charts Showing Flat Lines

The default forecast method was `"ema"`, which by design projects the last rolling average forward as a flat horizontal line. Switched to `"linear"` (OLS trend extrapolation) in `config.py`.

2. Date & Timezone Fix — Scraper

Modified `scraper.py`

The scraper was tagging data with `datetime.date.today()` — the local Tunisia date. Since Tunisia is UTC+1 and the user sometimes runs the scraper at 3 AM local time (still Sunday in the US), data scraped on Monday local was being tagged as Monday when Yahoo Finance was actually serving Friday's closing prices.

Fix

Added `get_us_market_date()` function using `zoneinfo.ZoneInfo("America/New_York")`. Returns the most recent US trading day date, rolling back to Friday on weekends. All data is now tagged with the correct US market date regardless of where the scraper is running.

3. Historical Data Extension

Problem

All intelligence modules (forecasting, signals, anomaly detection, AI briefing) were operating on only 4 days of scraped data. Z-scores, rolling averages, and baselines computed from 4 points are statistically meaningless.

Solution — Three Extended Data Builders

New Functions added to `historical_metrics.py`

Function	What it builds	Used by
`build_extended_hist_df()`	Market-wide daily aggregates: combines OHLCV-derived avg_pct / volatility / breadth_pct (500 days) with scraped data. Scraped dates override OHLCV on overlap.	Forecasting, AI Narratives
`build_extended_df_full()`	Per-symbol price rows: appends OHLCV-derived daily % Change for each symbol (434 days avg) before the scraper started.	Signal Engine, Anomaly Detection, Risk Scores, Opportunity Ranking
`_ohlcv_daily_aggregates()`	Internal helper: loads all 124 cached OHLCV parquet files, computes cross-sectional daily metrics.	Called by build_extended_hist_df()

> Result: 4 scraped days → 502-day baseline for forecasting. 4 scraped rows per symbol → 434 rows avg for signal z-scores. Data grows daily as scraping continues, gradually replacing OHLCV.

4. Historical Sentiment Extension

VIX as Market-Wide Sentiment Proxy

New `finnhub_sentiment.py` | Modified `sentiment.py`

Added `fetch_vix_history(period="2y")` which downloads VIX daily closes from yfinance and converts each reading to a VADER-compatible sentiment score using:

```
score = -tanh((VIX - 20) / 10)
```

VIX 10 → +0.76 (bullish) · VIX 20 → 0.00 (neutral) · VIX 30 → -0.76 (bearish)

Finnhub Historical News Sentiment

Added Finnhub's free `/company-news` API as a historical news source. Headlines are scored with VADER (same approach as the daily scraper). Rate-limited to 60 calls/min to respect the free tier.

Extended Daily Sentiment Average

Added `extended_daily_sentiment_avg()` to `sentiment.py`. Combines:

- Finnhub/VADER history (384 days backfilled in one-time run)
- VIX-derived scores for any remaining gaps
- Fresh daily VADER scores from the scraper for recent days

Result: 4-day sentiment baseline → 631-day extended baseline.

One-Time Backfill Scripts

Script	Purpose	Command
`backfill_sentiment.py`	Seed sentiment history from Finnhub	`python backfill_sentiment.py --from 2024-01-01`
`backfill_date.py`	Fill a missed scraping day from yfinance	`python backfill_date.py 2025-05-30`

> API key is stored in `.env` (gitignored). Scripts read it automatically — no `--key` argument needed after first setup.

5. Forecasting Improvements

Modified `config.py` | `dashboard_components/forecast_tab.py` | `dashboard.py`

- Changed `DEFAULT_METHOD` from `"ema"` (flat line) to `"linear"` (OLS trend extrapolation)
- Forecast computation now trains on 502-day extended hist_df instead of 4 scraped days
- Forecast charts now show **14 days of actual data** (dotted line) before the 5-day projection (solid line), making the trend direction visible
- Switched from category x-axis (one label per day, very cluttered) to date axis with **weekly ticks**
- Added dashed vertical separator at the today/forecast boundary
- Shaded 95% confidence band widening over the forecast period

6. Signal Intelligence Improvements

Modified `main.py` → `stage_signal_engine()`

All 4 per-symbol detectors (`detect_breakout`, `detect_unusual_movers`, `compute_risk_scores`, `compute_opportunity_ranking`) now receive `extended_df_full` instead of scraped-only `df_full`.

Signal Type	Before	After
momentum_breakout	0 detected (min history not met)	Fires with real 7-day rolling z-scores
unusual_mover	0 detected (min history not met)	Fires with 2-year per-symbol distribution
volatility_spike	Working (no history needed)	Unchanged
sentiment_divergence	Working (no history needed)	Unchanged

Signal count went from **25** → **39** after the extension. Z-scores like "DELL z=+7.91 vs full history (mean +0.32%, std 4.10%)" are now based on 2 years of real price data, not 4 days.

Anomaly Detection also receives `extended_df_full`, unlocking the `detect_price_gap_anomalies` detector which requires per-symbol history.

7. AI Briefing Improvements

Modified `main.py` → `stage_ai_narratives()`

Previously the AI briefing received the 4-day scraped `hist_df`, so commentary like "today's move is above the 7-day average" was comparing against near-meaningless baselines.

Now passes `extended_hist` (502 days) so:

- Session comparison is against a real 7-day rolling average grounded in market history
- Volatility baseline (`baseline_vol`) comes from the 500-day mean, not 4 points
- Commentary like "volatility is X% above historical norm" has actual statistical meaning

8. Task Scheduler — Fully Automated Daily Pipeline

New `run_daily.bat` | `run_daily.sh` | `setup_scheduler.py`

Platform	Method	Schedule
Windows (this PC)	Windows Task Scheduler via `run_daily.bat`	Mon–Fri at 22:30 local time (after US market close)
Linux / Cloud	cron job via `run_daily.sh --install`	`30 21 * * 1-5` (21:30 UTC)

> The pipeline runs fully automatically. No manual intervention needed. Logs go to `logs/scheduler.log`.

9. Auto-Backfill on Every Run

New `stage_auto_backfill()` in `main.py`

Added as the first stage of every `main.py` run. Detects and fills two classes of gaps silently:

- **Missing price days:** Scans the last 30 days for weekday gaps. If the PC was off or the scraper missed a day, fills automatically using yfinance closing prices.
- **Sparse sentiment:** If sentiment history has fallen more than 7 days behind the price data, auto-runs a Finnhub backfill for the missing period.

No manual scripts ever need to be run — the pipeline self-heals.

10. OHLCV Fetch Optimization

Modified `pipelines/fetch_historical.py`

Stage 0 (OHLCV fetch) was running on every `main.py` call, making 124+ yfinance API checks even when data was already current. Added a sentinel file (`.ohlcv_last_fetch`) that records the last US market date the fetch ran.

On subsequent runs the same day, the entire stage is skipped in milliseconds instead of running for ~60 seconds. The sentinel resets automatically the next trading day.

11. Complete File Change Log

File	Status	Summary
`scraper.py`	Modified	US Eastern time date tagging via `get_us_market_date()`
`historical_metrics.py`	Modified	Added `build_extended_hist_df`, `build_extended_df_full`, `_ohlcv_daily_aggregates`
`main.py`	Modified	Auto-backfill stage, extended data wired into all stages, DataFrame or-bug fixes
`sentiment.py`	Modified	Added `extended_daily_sentiment_avg()` blending VIX + VADER
`config.py`	Modified	Forecast method changed from `ema` to `linear`
`dashboard.py`	Modified	Passes `extended_hist_df` to forecast chart builder
`dashboard_components/forecast_tab.py`	Modified	14-day history context, date axis, weekly ticks, today separator
`pipelines/fetch_historical.py`	Modified	Sentinel file — skips OHLCV fetch if already run today
`backfill_date.py`	New	CLI to backfill a missed trading day from yfinance
`finnhub_sentiment.py`	New	Finnhub news + VIX historical sentiment integration
`backfill_sentiment.py`	New	CLI to seed sentiment history from Finnhub
`run_daily.bat`	New	Windows scheduled run script
`run_daily.sh`	New	Linux/cloud cron script with `--install` / `--remove`
`setup_scheduler.py`	New	Registers Windows Task Scheduler task programmatically
`.env`	New	Finnhub API key (gitignored)
`.gitignore`	New	Protects .env, parquet files, logs from git commits

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